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IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2026-27-I: SOLUTIONS
Date: August 25, 2026

CS 405/6001: Game Theory and Algorithmic Mechanism Design

Total: 40 marks, Duration: 1 hour, ATTEMPT ALL QUESTIONS

Instructions:

1. This question-and-answersheet booklet contains a total of 6 sheets of paper (12 pages, pages 2 and 12 are blank). Please verify.
2. Write your roll number and department on **every side of every sheet** (except the blank sheet) of this booklet. The first 5 minutes of additional time is given exclusively for this activity. Use only **black/blue ball-point pen**.
3. Write final answers neatly with a pen **only in the given boxes**.
4. Use the rough sheets for scratch works / attempts to solution. **Write only the final solution (which may be a sequence of logical arguments) in a precise and succinct manner in the boxes provided.** Do not provide unnecessarily elaborate steps. The space within the boxes are sufficient for the correct and precise answers.
5. Submit your answerscripts to the teaching staff when you leave the exam hall or the time runs out (whichever is earlier). **Your exam will not be graded if you fail to return the paper.**
6. **This is a closed book, notes, internet exam. No communication device, e.g., cellphones, iPad, smartwatches, etc., is allowed.** Keep them switched off in your bag and keep the bag away from you. If anyone is found in possession of such devices during the exam, that answerscript may be disqualified for evaluation and DADAC may be invoked.
7. One A4 assistance sheet (text **only on one side**) is allowed for the exam.
8. **After you are done with your exam or the exam duration is over, please DO NOT rush to the desk for submitting your paper.** Please remain seated until we collect all the papers, count them, and give a clearance to leave your seat.

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Problem 1 (8 points). The Complementary Halving Game. Players I and II alternate turns, Player I moving first. The state of the game is a single nonnegative integer n . A move replaces the current n by either $n - 1$ (Decrement) or $\lfloor n/2 \rfloor$ (Halve). A player **wins** the game whenever he moves the state to $n = 0$; equivalently, the player whose turn it is to move when $n = 0$ loses. So, unlike the halving game discussed in class, in which the player who moves the state to 0 is the one who *loses*, here the two players race to *reach* 0, hence the name *complementary*.

Recall also that a strategy of a player is a **winning strategy** if it guarantees that player a win against **every** strategy of the opponent, in particular against the opponent's best attempt to defeat him.

- (a) Call a state n a **W**-state if the player whose turn it is to move at n has a winning strategy, and an **L**-state otherwise. Fill in the table below (the entry for $n = 0$ is filled in for you), and then state which player has a winning strategy when the game starts at $n = 8$. **1 + 1 = 2 points.**

n	0	1	2	3	4	5	6	7	8
W or L	L	W	L	W	W	W	L	W	L

Starting at $n = 8$, the player with a winning strategy is **Player II**.

(Marking note: a state n is a **W**-state exactly when at least one of its two children $n - 1$, $\lfloor n/2 \rfloor$ is an **L**-state. Working upward from $n = 0$ (**L**, given): $1 \rightarrow \{0, 0\}$: **W**; $2 \rightarrow \{1, 1\}$: **L**; $3 \rightarrow \{2, 1\}$: **W**; $4 \rightarrow \{3, 2\}$: **W**; $5 \rightarrow \{4, 2\}$: **W**; $6 \rightarrow \{5, 3\}$: **L**; $7 \rightarrow \{6, 3\}$: **W**; $8 \rightarrow \{7, 4\}$: **L**. As $n = 8$ is an **L**-state and Player I moves first, Player II wins. No written justification is asked for in this part.)

- (b) Recall the following result due to von-Neumann from the class.

Theorem. Let Γ be a two-player game in which (C1) the game tree is finite, (C2) there are no chance moves, and (C3) the players have perfect information, and whose set of outcomes is $O = \{\text{I wins, II wins, Draw}\}$. Then exactly one of the following three alternatives holds: (i) Player I has a winning strategy; (ii) Player II has a winning strategy; (iii) each of the two players has a strategy guaranteeing at least a draw.

Is the following statement TRUE or FALSE? “The Theorem above applies to the Complementary Halving Game started at any n , and there exist starting values of n for which alternative (iii) is the one that holds.” **1 point.**

FALSE

- (c) Justify your answer to Part (b): if your answer there is TRUE, give a formal proof; if it is FALSE, give the precise reason for its failure. If your answer to the previous part is incorrect, this part will not be graded. **2 points.**

The Theorem does apply: n strictly decreases at every move, so the tree is finite (C1); the moves Decrement and Halve are deterministic, so there are no chance moves (C2); each player sees the current n and the entire history (C3). But the Complementary Halving Game has **no draw outcome**: every play terminates at $n = 0$ with exactly one player unable to move, hence with a winner. So alternative (iii), which requires a guaranteed win or draw for both players simultaneously, can never hold; exactly one of (i), (ii) holds for every starting n .

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- (d) **The blue-eyed islanders, again.** On an isolated island with no mirrors and no discussion of eye color live n islanders, exactly k of whom have blue eyes. Everyone can see everyone else's eyes but not their own, everyone reasons perfectly, and this is common knowledge. Any islander who deduces that his own eyes are blue must leave that same night.

A visiting outsider, known to always tell the truth, announces publicly and simultaneously to all of them: "*At least two of you have blue eyes.*" Assume $k \geq 2$, so the announcement is true. On which night do all the blue-eyed islanders leave? **1 point.**

All blue-eyed islanders leave on night $k - 1$.

- (e) Justify your answer to Part (d) with a precise argument. *If your answer to the previous part is incorrect, this part will not be graded.* **2 points.**

Induction on k . Base case $k = 2$: each of the two blue-eyed islanders sees exactly one blue-eyed islander, and the announcement says there are at least two, so the second one must be himself. Both leave on night $1 = k - 1$.

Inductive step: suppose the claim holds for $k - 1 \geq 2$. With k blue-eyed islanders, each of them sees exactly $k - 1$ blue-eyed islanders. Had there been only $k - 1$, those $k - 1$ would all have left on night $k - 2$ by the inductive hypothesis. Night $k - 2$ passes with nobody leaving, so each blue-eyed islander concludes there is a k -th blue-eyed islander besides the $k - 1$ he sees, namely himself, and all k leave on night $k - 1$. (No black-eyed islander can ever conclude anything about his own eyes, so nobody else leaves.)

What the announcement does is not to add raw information, since everyone could already see at least two blue-eyed islanders when $k \geq 3$, but to make the fact *common knowledge*, which is what lets the induction start. Announcing "at least two" shifts the base case by one night relative to the "at least one" version, whose answer is night k .

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Problem 2 (9 points). Consider the following two-player game G , where Player 1 picks a row and Player 2 picks a column.

		Player 2		
		L	C	R
Player 1	T	4, 2	2, 3	2, 1
	M	3, 2	3, 3	1, 4
	B	2, 2	2, 2	0, 5

Recall that player i 's **maxmin (security) value** is $\underline{v}_i = \max_{s_i \in S_i} \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$, and any maximizer is a **maxmin strategy** of i .

- (a) For each player, list every strategy that is **strictly** dominated, and every strategy that is **weakly but not strictly** dominated, in G . Throughout this part, domination is by a **pure** strategy (and not by a mixed strategy). In each case name a strategy that does the domination. Write “none” where applicable. **1 + 1 = 2 points.**

Player 1: strictly dominated: B , by M (since $3 > 2$, $3 > 2$, $1 > 0$). Weakly but not strictly dominated: none. (T also weakly dominates B , as $4 > 2$, $2 = 2$, $2 > 0$, but B is already strictly dominated; T and M do not dominate each other, since $4 > 3$ but $2 < 3$.)

Player 2: strictly dominated: none. Weakly but not strictly dominated: L , by C (Player 2's payoffs are $L = (2, 2, 2)$ and $C = (3, 3, 2)$ down the rows T, M, B : strictly better in rows T and M , tied in row B). $R = (1, 4, 5)$ is not dominated by anything.

- (b) Compute the maxmin values $\underline{v}_1$ and $\underline{v}_2$ in G , and list all maxmin strategies of each player.

1 + 1 = 2 points.

Row minima of Player 1's payoffs: $T : \min(4, 2, 2) = 2$; $M : \min(3, 3, 1) = 1$; $B : \min(2, 2, 0) = 0$. Hence $\underline{v}_1 = 2$, with the unique maxmin strategy T .

Column minima of Player 2's payoffs (taken down the rows): $L : \min(2, 2, 2) = 2$; $C : \min(3, 3, 2) = 2$; $R : \min(1, 4, 5) = 1$. Hence $\underline{v}_2 = 2$, with maxmin strategies L and C (both attain 2).

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- (c) Is the following statement TRUE or FALSE? “In every strategic form game, deleting a strictly dominated strategy of Player 1 leaves Player 2’s maxmin value unchanged.”

1 point.

FALSE

- (d) Justify your answer to Part (c): if your answer there is TRUE, give a formal proof; if it is FALSE, give a small counterexample. *If your answer to the previous part is incorrect, this part will not be graded.* You may use the game G above. **2 points.**

G itself is a counterexample. Deleting Player 1’s strictly dominated row B leaves Player 2’s column minima as $L : \min(2, 2) = 2$, $C : \min(3, 3) = 3$, $R : \min(1, 4) = 1$, so v_2 rises from 2 to **3** (with C now the unique maxmin strategy). Player 2 had to guard against row B only because it was available; once it is gone, the worst case that dragged C down to 2 disappears. (What *is* true is that deleting one’s **own** dominated strategy leaves one’s own maxmin value unchanged, and that another player’s deletion can only raise it.)

- (e) Does G have a pure strategy Nash equilibrium (PSNE)? If yes, list all of them; if no, prove it **without checking all nine cells** of G , by using elimination arguments. **2 points.**

No, G has no PSNE.

Elimination of a *strictly* dominated strategy neither destroys a PSNE nor creates a new one, so G and the reduced game have exactly the same PSNE. Delete Player 1’s strictly dominated B ; in the resulting 2×3 game Player 2’s payoffs are $L = (2, 2)$ and $C = (3, 3)$, so now C *strictly* dominates L , and deleting L is again PSNE-preserving. What remains is

		Player 2	
		C	R
Player 1	T	2, 3	2, 1
	M	3, 3	1, 4

Here Player 1’s best responses are M to C and T to R , while Player 2’s best responses are C to T and R to M . The two best-response maps never agree, so the 2×2 game has no PSNE, and therefore neither does G .

(Note that using *weak* domination to eliminate L from G directly would not have been a valid argument: weak elimination can destroy a PSNE of the original game.)

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Problem 3 (10 points). (a) The matrix below gives **Player 1's** payoffs in a two-player zero-sum game (Player 2's payoffs are the negatives of these entries).

		Player 2		
		<i>L</i>	<i>C</i>	<i>R</i>
Player 1	<i>T</i>	4	-1	2
	<i>M</i>	3	1	5
	<i>B</i>	0	-3	1

Recall $\underline{v} = \max_{s_1} \min_{s_2} u(s_1, s_2)$ and $\bar{v} = \min_{s_2} \max_{s_1} u(s_1, s_2)$, and that (s_1^*, s_2^*) is a **saddle point** of u if $u(s_1^*, s_2^*)$ is simultaneously the maximum of its column and the minimum of its row. Fill in the blanks. Write “none” where applicable. **$0.5 \times 6 = 3$ points.**

$\underline{v} =$ 1 maxmin strategies of Player 1: *M*

$\bar{v} =$ 1 minmax strategies of Player 2: *C*

all saddle points: (M, C) value of the game: 1

(Marking note: row minima: $T : \min(4, -1, 2) = -1$, $M : \min(3, 1, 5) = 1$, $B : \min(0, -3, 1) = -3$, so $\underline{v} = 1$, attained uniquely at M . Column maxima: $L : 4$, $C : 1$, $R : 5$, so $\bar{v} = \min(4, 1, 5) = 1$, attained uniquely at C . Since $\underline{v} = \bar{v} = 1 = u(M, C)$, the cell (M, C) is the unique saddle point, equivalently the unique PSNE, and the value is 1.)

(b) The game below has exactly two pure strategy Nash equilibria, namely (C, C) and (F, F) . Find its **completely mixed** Nash equilibrium, i.e., the MSNE in which each player randomizes over both of her strategies, and give the expected payoff of each player in it. Show the indifference conditions you use. **$2 + 1 = 3$ points.**

		Player 2	
		<i>C</i>	<i>F</i>
Player 1	<i>C</i>	3, 1	0, 0
	<i>F</i>	0, 0	1, 2

Write $p = \sigma_1(C)$ and $q = \sigma_2(C)$, both strictly between 0 and 1. Since the support of each player is all of $\{C, F\}$, condition (C1) of the characterization theorem requires each player to be indifferent between her two pure strategies:

$$\begin{aligned} u_1(C, \sigma_2) &= u_1(F, \sigma_2) \implies 3q = 1 - q \implies q = \frac{1}{4}, \\ u_2(\sigma_1, C) &= u_2(\sigma_1, F) \implies p = 2(1 - p) \implies p = \frac{2}{3}. \end{aligned}$$

Condition (C2) is vacuous, as no pure strategy lies outside the support. Hence the completely mixed MSNE is $((\frac{2}{3}, \frac{1}{3}), (\frac{1}{4}, \frac{3}{4}))$, with expected payoffs $u_1 = 3q = \frac{3}{4}$ and $u_2 = p = \frac{2}{3}$, i.e., the profile $(\frac{3}{4}, \frac{2}{3})$, which is worse for both players than either pure equilibrium.

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- (c) In the class we stated, but did not prove, the following result. Prove it. You may use the MSNE characterization theorem, which says that $(\sigma_i^*, \sigma_{-i}^*)$ is an MSNE if and only if for every player i : (C1) $u_i(s_i, \sigma_{-i}^*)$ takes the same value for every $s_i \in \delta(\sigma_i^*)$, and (C2) $u_i(s_i, \sigma_{-i}^*) \geq u_i(s'_i, \sigma_{-i}^*)$ for every $s_i \in \delta(\sigma_i^*)$ and every $s'_i \notin \delta(\sigma_i^*)$. Your proof should not need more than three or four steps. **4 points.**

Theorem. Let $s_i \in S_i$ be strictly dominated by some mixed strategy $\sigma_i \in \Delta S_i$, i.e., $u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i})$ for every $s_{-i} \in S_{-i}$. Then in every MSNE of the game, s_i is played with probability zero.

Suppose, for contradiction, that $(\sigma_i^*, \sigma_{-i}^*)$ is an MSNE in which s_i lies in the support $\delta(\sigma_i^*)$ of player i 's equilibrium strategy.

Step 1 (strict domination survives against a mixed opponent profile). Expected utility is linear in the opponents' mixed profile, so

$$u_i(\sigma_i, \sigma_{-i}^*) = \sum_{s_{-i} \in S_{-i}} \sigma_{-i}^*(s_{-i}) u_i(\sigma_i, s_{-i}) > \sum_{s_{-i} \in S_{-i}} \sigma_{-i}^*(s_{-i}) u_i(s_i, s_{-i}) = u_i(s_i, \sigma_{-i}^*),$$

the strict inequality holding because the weights $\sigma_{-i}^*(s_{-i})$ are non-negative and sum to 1 (so at least one is strictly positive) while the inequality $u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i})$ holds termwise.

Step 2 (indifference on the support). Since $s_i \in \delta(\sigma_i^*)$, condition (C1) makes $u_i(s'_i, \sigma_{-i}^*)$ the same for every $s'_i \in \delta(\sigma_i^*)$, hence

$$u_i(\sigma_i^*, \sigma_{-i}^*) = \sum_{s'_i \in \delta(\sigma_i^*)} \sigma_i^*(s'_i) u_i(s'_i, \sigma_{-i}^*) = u_i(s_i, \sigma_{-i}^*).$$

Step 3 (the contradiction, directly). Combining the two steps,

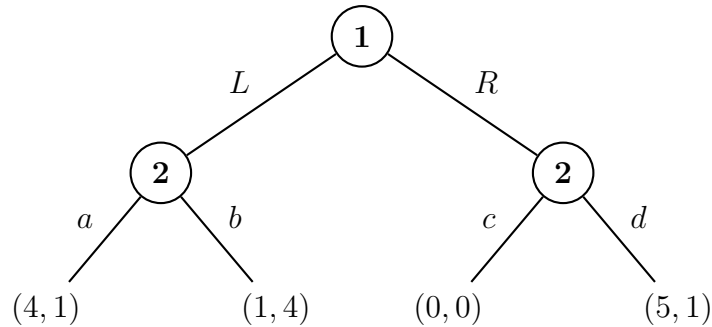
$$u_i(\sigma_i, \sigma_{-i}^*) > u_i(s_i, \sigma_{-i}^*) = u_i(\sigma_i^*, \sigma_{-i}^*),$$

i.e., against σ_{-i}^* player i gets strictly more from the mixed strategy σ_i than from σ_i^* . That is a profitable unilateral deviation, contradicting the assumption that $(\sigma_i^*, \sigma_{-i}^*)$ is an MSNE. Hence $s_i \notin \delta(\sigma_i^*)$, i.e., $\sigma_i^*(s_i) = 0$ in every MSNE. ■

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Problem 4 (13 points). Consider the perfect information extensive form game (PIEFG) below. Circled labels are the player who moves at that history, edge labels are actions, and the leaves carry the payoff pair (u_1, u_2) .



- (a) Write down the pure strategy sets S_1 and S_2 , and then fill in the normal form of this game in the grid below (write Player 2's strategies along the top row and Player 1's strategies down the first column).

1 + 3 = 4 points.

Player 1 moves only at the root, so $S_1 = \{L, R\}$. Player 2 moves at the two histories (L) , with actions $\{a, b\}$, and (R) , with actions $\{c, d\}$, so a strategy of hers is a pair, $S_2 = \{ac, ad, bc, bd\}$: it fixes an action even at the history that Player 1's move makes unreachable. That is why the columns ac, bc agree on the row R , and ad, bd agree on the row R as well.

		Player 2			
		ac	ad	bc	bd
Player 1	L	4, 1	4, 1	1, 4	1, 4
	R	0, 0	5, 1	0, 0	5, 1

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(b) List all the pure strategy Nash equilibria (PSNE) of this game.

3 points.**The PSNE are (L, bc) , (R, ad) and (R, bd) .**

Checking column by column. In column ac , Player 1's maximum is 4 at L , but at row L Player 2 prefers bc or bd (payoff $4 > 1$), so neither cell of ac is a PSNE, and the same argument rules out (L, ad) and (L, bd) . At (R, ad) , Player 1's 5 is the column maximum and Player 2's 1 is the row- R maximum: PSNE. In column bc , Player 1's maximum is 1 at L , and Player 2's row- L maximum 4 is attained at bc : PSNE. In column bd , Player 1's maximum is 5 at R , and Player 2's row- R maximum 1 is attained at bd : PSNE. The remaining cells, (R, ac) and (R, bc) , both pay Player 1 zero, and she gains by switching to L .

(c) Find the subgame perfect Nash equilibrium (SPNE) by backward induction. For each PSNE of Part (b) that is **not** an SPNE, name the subgame at which the profile prescribes a suboptimal action, i.e., the non-credible threat it rests on. **1 + 1 = 2 points.**

Backward induction. At Player 2's node after L : a pays her 1, b pays 4, so b , and the node is worth (1, 4). At Player 2's node after R : c pays her 0, d pays 1, so d , and the node is worth (5, 1). At the root, Player 1 compares 1 from L with 5 from R , so R .

Unique SPNE: (R, bd) , with outcome (5, 1) along the path R, d .

Non-credible threats. (L, bc) prescribes c in the subgame rooted at (R) , where c pays Player 2 0 against 1 from d : her threat to play c is exactly what deters Player 1 from choosing R , and it is not credible. (R, ad) prescribes a in the subgame rooted at (L) , where a pays Player 2 1 against 4 from b ; that subgame is off the equilibrium path, so the suboptimality is never exposed.

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- (d) Recall that a correlated strategy is a distribution π on $S_1 \times \cdots \times S_n$, and that π is a **correlated equilibrium (CE)** if for every player i and all $s_i, s'_i \in S_i$,

$$\sum_{s_{-i} \in S_{-i}} \pi(s_i, s_{-i}) u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} \pi(s_i, s_{-i}) u_i(s'_i, s_{-i}).$$

A mediator proposes the correlated strategy

$$\pi(L, bd) = \frac{1}{2}, \quad \pi(R, bd) = \frac{1}{2}, \quad \pi(s) = 0 \text{ for every other } s \in S_1 \times S_2.$$

Is π a correlated equilibrium of this game? Answer YES or NO.

2 points.

NO

- (e) Justify your answer to Part (d): if your answer there is YES, prove that no player gains by deviating from any recommendation she may receive; if it is NO, exhibit a profitable deviation of some player from some recommendation, writing out the corresponding inequality of the definition with all the numbers filled in. *If your answer to the previous part is incorrect, this part will not be graded.*

2 points.

Player 1 has a profitable deviation from the recommendation L . Only $s_2 = bd$ has $\pi(L, s_2) > 0$, so with $s_1 = L$ and $s'_1 = R$ the two sides of the CE inequality are

$$\text{comply: } \sum_{s_2} \pi(L, s_2) u_1(L, s_2) = \frac{1}{2} \cdot u_1(L, bd) = \frac{1}{2} \cdot 1 = \frac{1}{2},$$

$$\text{deviate: } \sum_{s_2} \pi(L, s_2) u_1(R, s_2) = \frac{1}{2} \cdot u_1(R, bd) = \frac{1}{2} \cdot 5 = \frac{5}{2},$$

and $\frac{1}{2} \not\geq \frac{5}{2}$. Told to play L , Player 1 infers that Player 2 has been told bd , and switching to R raises her payoff from 1 to 5.

(Player 2's own constraint does hold: she is always told bd , against which Player 1 plays L or R with probability $\frac{1}{2}$ each, so complying gives $\frac{1}{2}(4) + \frac{1}{2}(1) = \frac{5}{2}$, against $\frac{1}{2}$ from ac , 1 from ad and 2 from bc . The single failure above is enough. The root cause is that (L, bd) is not a PSNE of the game: a mediator who randomizes over profiles must, on each recommendation, leave the recommended player with no profitable unilateral override.)

END OF QUESTION PAPER. GOOD LUCK!

